

Personal Profile

Alexander Lange, Ph.D.,
High-Energy Astrophysics Researcher,
View my GitHub here,

Education

- 2021 - Present **Ph.D.,** *Department of Physics, The George Washington University*
- 2019 - 2023 **M.Phil.,** *Department of Physics, The George Washington University*
- 2021 - 2023 **Certificate in Data Mining,** *Department of Data Science, The George Washington University*
- 2019 - 2021 **MS,** *Department of Physics, The George Washington University*
- 2013 - 2017 **BA,** *Department of Physics and Astronomy, Vassar College*

Fellowships and Awards

- 2025 The George Washington Dissertation Completion Fellowship
- 2024 - 2025 Bell Atlantic and an Andrew Knox Endowment Fellowship
- 2020 Bennhold Family Award for Best Teaching on the Part of a Graduate Student

Research Assistantship

- Fall 2020 - Present Graduate Research Assistant, *NASA-GSFC / UMBC / CRESST II*
- Fall 2023 - Present Department of Physics, Graduate Research Assistant, *The George Washington University*

Research Publications (First Author)

- 2025 Lange et al., Published Jun. 2025. **Broadband modeling and analysis of High-Mass Gamma-Ray Binary: 4FGL J1405.1-6119.** doi: 10.3847/1538-4357/adda32
- 2025 Lange et al., Published Jul. 2025. **The Vela Pulsar and Pulsar Wind Nebula with 13 years of Fermi–LAT Observations.** doi: 10.3847/1538-4357/ade713

- 2025 Alexander Lange and B.B. K., Accepted Nov. 2025. **FermiPhased: A tool for phase-resolved likelihood analysis of Fermi-LAT data.** doi: <https://arxiv.org/pdf/2511.04810>
- In Prep. Lange et al., **NICER Spectral Timing Observations of Cyg X-3 During the 2018 Quenched/Hypersoft State.**

Research Publications (Contributing Author)

- 2025 Romano et al., Published Jul. 2025. **Swift/XRT monitoring of the orbital and superorbital modulations in 4U 1909+07.** doi: 10.1051/0004-6361/202554601
- 2024 Corbet et al., Published Nov. 2024. **Sharp Periodic Flares and Long-term Variability in the High-mass X-Ray Binary XTE J1829-098 from RXTE PCA, Swift BAT, and MAXI Observations.** doi: 10.3847/1538-4357/ad83b9
- 2024 Kargaltsev et al., Published Nov. 2024. **Hard X-Ray Spectrum of the Vela Pulsar and Its Wind Nebula Constrained by NuSTAR.** doi: 10.3847/1538-4357/ad7827
- 2024 Gagnon et al., Published Jun. 2024. **Chandra X-Ray Observations of PSR J1849-0001, Its Pulsar Wind Nebula, and the TeV Source HESS J1849-000.** doi: 10.3847/1538-4357/ad3e6d
- 2023 Kargaltsev et al., Published Nov. 2023. **A lack of 9-s periodicity in the follow-up NuSTAR observation of LS 5039.** doi: 10.3847/1538-4357/ad022d

Grant Proposals

NuSTAR **Primary Investigator of a Nuclear Spectroscopic Telescope Array (NuSTAR) Cycle 11; observation proposal. Observations of the high-mass gamma-ray binary Approved HESS J0632+057 for 130 ks with the NuSTAR space telescope. With the Category C observation comes a grant at a flat rate to supplement publication costs.**

Conferences and Oral Presentations

- August, 2025 **Host**
Astronomy-Consortium in D.C. (AC/DC) Graduate Student Conference 2025. *Washington, D.C.*
- August, 2025 **Presenter**
Phase-Resolved Analysis of Fermi—LAT data with a Fermi GUI and test cases: 4FGL J1405.1-6119, The Vela Pulsar and 4FGL J1702.7-5655. AC/DC Graduate Student Conference 2025. *Washington, D.C., George Washington University*
- June, 2025 **Presenter**
Phase-Resolved Analysis of Fermi—LAT data with a Fermi GUI and test cases: 4FGL J1405.1-6119, The Vela Pulsar and 4FGL J1702.7-5655. Fermi Summer School 2025. *Lewes, University of Delaware*

- October, **Presenter**
 2024 Orbital phase-resolved analysis of X-ray and Gamma-ray observations of High-Mass Gamma-ray Binary 4FGL J1405.1-6119. DIAS-CDY Workshop - Gamma-Ray Loud Binaries. *Dublin, Ireland*
- September, **Poster**
 2024 Orbital phase-resolved analysis of X-ray and Gamma-ray observations of High-Mass Gamma-ray Binary 4FGL J1405.1-6119. 11th National Fermi Symposium. *College Park, University of Maryland*
- September, **Oral Session**
 2024 The Vela pulsar and its pulsar wind nebula Vela-X using 13 years of Fermi-LAT Observations. 11th International Fermi Symposium. *College Park, University of Maryland*
- April, 2024 **Poster**
 A NICER and SMA Study of the high-mass gamma-ray binary Cygnus X-3 as a function of binary orbit and spectral state. 21th High Energy Astrophysics Division Meeting. *Horseshoe Bay, Texas*
- August, 2024 **Host**
 Astronomy-Consortium in D.C. (AC/DC) Graduate Student Conference. *Washington, D.C.*
- August, 2024 **Oral Session**
 The Vela pulsar and its pulsar wind nebula Vela-X using 13 years of Fermi-LAT Observations. 11th International Fermi Symposium. *Washington, D.C.*
- April, 2024 **Poster**
 Two NuSTAR observations of gamma-ray binary LS 5039: pulsation search and spectral variations with the orbital phase. 21th High Energy Astrophysics Division Meeting. *Horseshoe Bay, Texas*
- October, **Oral Session**
 2023 Vela Pulsar and Vela-X: An update on the multi-wavelength picture of Vela and its PWN. *University of Maryland, College Park, Department of Physics and Astronomy*
- September, **Oral Session**
 2023 Writing with ChatGPT: Usage of Popular Neural Network Models in WID. *George Washington University's Writing Department, Washington, D.C.*
- August, 2023 **Host**
 DMV Astrophysics Consortium Student Conference. *Washington, D.C.*
- August, 2023 **Poster**
 A review of Gamma-ray Binaries, their emission mechanisms and orbital properties. *DMV Astrophysics Consortium Student Conference. Washington, D.C.*
- August, 2023 **Oral Session**
 A Case Study of High-Mass Gamma-Ray Binaries: 4FGL J1405.1-6119. *DMV Astrophysics Consortium Student Conference. Washington, D.C.*
- March, 2023 **Poster**
 High-Mass Gamma-Ray Binary: J1405.1-6119. 19th *Kuila-Kona, Hawai'i*

March, 2023 **Poster**
Vela pulsar from hard X-rays to GeV Gamma-Rays. 19th Kuilau-Kona, Hawai'i

News Publications

Feb, 2024 **Blog Post**
DC-Area Astrophysics Graduate Student Conference Revived! *American Astronomy Society, Education Committee Blog* <https://aas.org/posts/news/2024/01/dc-area-astrophysics-graduate-student-conference-revived>

Graduate Teaching Assistantships

Fall 2020 Graduate Assistant, *The George Washington University, ASTR 1001*
Spring 2021 Graduate Assistant, *The George Washington University, Department of Physics, PHYS 1012*
Summer 2021 Graduate Assistant, *The George Washington University, Department of Physics, PHYS 1021*
Summer 2021 Graduate Assistant, *The George Washington University, Department of Physics, PHYS 1022*
Fall 2021 Graduate Assistant, *The George Washington University, Writing in the Disciplines, EMSE 3740W*
Spring 2022 Graduate Assistant, *The George Washington University, Writing in the Disciplines, MAE 4152W*
Fall 2022 Graduate Assistant, *The George Washington University, Writing in the Disciplines, EMSE 3740W*
Spring 2023 Graduate Assistant, *The George Washington University, Writing in the Disciplines, MAE 4152W*

Career Summary

Alex Lange earned his undergraduate degree from Vassar College and a PhD in Physics at George Washington University in 2025. After completing an internship at NASA Goddard Space Flight Center in the Guidance, Navigation & Control (GN&C) branch (Code 596), he spent a year teaching highschool physics and math in New York City for a charter school before transitioning back to academia.

Alex specializes in high-energy astrophysics, now working as a graduate researcher in Code 660, with a focus on complex systems such as High-mass X-ray Binaries, Gamma-ray Binaries, and Pulsars. He has authored two peer-reviewed, high-impact publications and contributed to multiple others, using statistical inference, time-series analysis, and multi-wavelength data integration. His research involves extracting weak signals from noisy data, modeling physical systems under uncertainty, and developing reproducible analysis pipelines—skills that translate directly to quantitative finance.

Alex has extensive expertise with high-energy astrophysics analysis tools, particularly `FermiTools` and `FermiPy`, which he uses for LAT data analysis and modeling. He is also proficient in `XSpec` for X-ray spectral fitting and `PINT` for high-precision pulsar timing, with prior experience using `Tempo`. Alex is well-versed in CIAO tools for Chandra data reduction and maintains strong version control practices through `GitHub`. His scripting skills span `bash`, `C`, and `perl`, with additional experience in creative and development platforms like `Blender` (for animations) and `Xcode`. He has also developed an interactive Python-based GUI application, `FermiPhased`, that automates phase-resolved analysis of Fermi-LAT data and spectral modeling into a seamless, user-friendly workflow for high-energy astrophysics.

Alex has been awarded several competitive fellowships and research grants in recognition of his academic excellence and research contributions. He was selected as an Atlantic Bell Graduate Fellow, a merit-based award supporting outstanding doctoral students at George Washington University. In his final year, he was also granted a Dissertation Writing Fellowship to support the completion of his PhD research and writing. Additionally, Alex secured a NASA observing grant as Principal Investigator, enabling the acquisition and analysis of space-based X-ray observations critical to his dissertation work on compact objects and high-energy astrophysical systems.

Driven by a deep interest in data modeling and computation, Alex is focused on developing advanced analytical tools and intelligent systems across diverse applications for all types of data. He designs and implements software in Python, C, and R, whether he is building end-to-end applications that integrate statistical modeling, signal processing, or neural network-driven classification using frameworks such as `TensorFlow`, `Keras`, and `scikit-learn`. His work includes classifiers, Monte Carlo Markov Chain simulation analysis, and Bayesian inference models for complex, noisy datasets. Through this, he combines rigorous statistical modeling with high-performance computing, programming and automation, applying the same attention to detail and intuition that were paramount in his years of astrophysical research and data analysis.